17 The first law of thermodynamics may be expressed as

$$\Delta U = Q + W$$

where ΔU is the increase in energy of the system, Q is the thermal energy supplied to the system and W is the work done on the system.

Which of the following is correct for the case of an isothermal expansion of an ideal gas?

	ΔU	Q	W
A	zero	positive	negative
B	positive	zero	positive
C	zero	negative	positive
D	negative	zero	negative

Ans: A

Increase in internal energy, ΔU is **zero** for an *isothermal* process.

Work done on gas, $W = -p\Delta V$ is **negative** ΔV is postive for an *expansion*.

By First Law of Thermodynamics, $\Delta U = Q + W$, W must be **positive**.